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Unlocking Potential: Predictive Analytics in Football

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1 Introduction

1.1 Business Problem

1.1.1 Optimizing Football Player Positions

Football clubs at every level face a persistent challenge: figuring out where a player will genuinely perform best. A club can invest significantly in a player only to see performances fall flat because the player has been placed in the wrong role — a problem that is more common than most clubs would publicly admit. NAC Breda, as the client for this project, operates in a competitive league environment where every squad decision carries weight.

1.1.2 Balancing Risk and Data in Player Positioning

Existing datasets capture where a player has performed in the past, but clubs are cautious about experimenting with new positions mid-season. With thirty-plus league games a year, there is almost no room to trial a player in an unfamiliar role during a high-stakes match. What clubs need is a way to test positional hypotheses without putting them into practice on the pitch until there is real evidence it will work.

1.1.3 Predictive Models for Player Performance

The answer is a predictive model that can anticipate, from a player's statistics alone, which broad position they are most suited to. Rather than relying on gut feeling or simply repeating what a player has always done, a data-driven model provides an objective second opinion. It does not replace coaching staff judgement, but it gives them something concrete to work with.

1.1.4 Revolutionizing Player Position Allocation

Without a reliable predictive tool, clubs default to conservative choices and miss potentially transformative positional moves. This project is built around the idea that machine learning can close that gap — providing a system that analyses player statistics across 114 features and outputs a well-grounded prediction of the position category where that player will have the most impact.

1.2 Solution

1.2.1 Transforming Player Positioning with Advanced Predictive Modelling

The solution developed here is a multi-class classification pipeline trained on a dataset of European football player statistics. It processes features including physical attributes, passing metrics, defensive contributions, attacking output, and positional tendencies, then predicts whether a player's optimal broad category is forward, midfielder, defender, or goalkeeper.

Multiple models were trained and compared, with the Random Forest consistently delivering the strongest results on this task.

1.2.2 Predictive Analysis for Optimal Player Positions

Beyond position prediction, the project also includes regression modelling to predict market value and expected assists (xA per 90), a MySQL database pipeline for persistent data storage, and a Streamlit application for interactive visualisation of team market value data. Together these components form a complete data science workflow from raw data to deployable tool.

1.2.3 Data-Driven Evolution in Football Strategy

This project aligns with the broader industry trend of embracing data-driven insights to gain a competitive edge. The implementation of a predictive model is not merely a technical exercise but a strategic tool — one that gives NAC Breda's scouting and analysis team a faster, more objective starting point for player evaluation.

2 Exploratory Data Analysis

2.1 Introduction

The EDA phase had a clear purpose: understand the data before building anything with it. The dataset contains player statistics from multiple European leagues, with 114 columns covering every aspect of player performance. The EDA was designed to reveal how different statistical features vary across positions, where missing data is concentrated, and which patterns would inform feature selection and model design.

2.2 Dataset Overview

The dataset was loaded from a CSV file (copy_data.csv) and contains records for professional football players across multiple European leagues and divisions. It has 114 columns, identified in the notebook as follows:

- Categorical columns (9): Player, Team, Team within selected timeframe, Position, Contract expires, Birth country, Passport country, Foot, On loan
- Numerical columns (105): Age, Market value, Matches played, Minutes played, Goals, xG, Assists, xA, and 97 further per-90 and percentage-based statistical metrics covering attacking, passing, defensive, set piece, and goalkeeper categories

The notebook confirmed the column names by printing data.columns.tolist(), which returned all 114 columns including goalkeeper-specific metrics (Save rate, xG against, Prevented goals) alongside outfield statistics.

2.3 Data Preparation

The data preparation steps documented in the notebook included:

- Loading the data with pd.read_csv() from a local file path.
- Identifying categorical and numerical columns using dtype detection.
- Handling missing values in the Age column by filling with the column mean (mean = 25.23).
- Replacing missing values in Birth country and Passport country with 'unknown'.
- Replacing missing values in the Foot column with 'unknown'.
- Converting the Contract expires column to datetime and calculating Contract Duration Left in years.
- Applying StandardScaler to numerical features and OneHotEncoder to categorical features via a ColumnTransformer pipeline.
- Using SimpleImputer with mean strategy for numerical features and most_frequent for categorical features within the modelling pipelines.

- Applying TruncatedSVD for dimensionality reduction (500 components) before the regression models.

2.4 Visual Techniques

The following visualisation functions were written and called in the notebook:

- Bar chart: average number of matches played by players in different age groups (age bins: 18–25, 25–30, 30–35, 35–40, 40–45), using sns.barplot with Seaborn.
- Bar chart: successful defensive actions per 90 split by on-loan vs permanent players, with standard deviation error bars.
- Bar chart: top 3 teams by average market value, used in the Streamlit application.
- Scatter plot: Age vs Market value (referenced in the report as Figure 2).
- Bar chart: average matches played by age group (Figure 3).
- Scatter and line plots: gradient descent cost over iterations and fitted regression line (Size vs Price).

All plots were built with Matplotlib and Seaborn following PEP 8 naming and commenting conventions throughout the notebook.

2.5 Variable Relationships

Correlation analysis and direct statistical queries in the notebook revealed the following:

- Height vs Goals: Pearson correlation = 0.06 — negligible relationship between physical size and goal-scoring.
- Weight vs Goals: Pearson correlation = 0.06 — same result; physical weight is not a useful predictor of attacking output.
- Italy was the most represented birth country in the dataset across all teams.
- Players aged 21–25 and 25–30 had the highest average matches played, consistent with professional peak years.
- The player with the highest xG in the dataset was M. Dugandžić (xG = 23.23, actual Goals = 22) — a near-perfect conversion rate.
- Contract Duration Left was calculated per team; some teams showed negative values, indicating expired contracts in the dataset.
- Duels won % and Aerial duels won % were calculated across all 1,533 unique position combinations, showing variation particularly at position extremes.

2.6 Key Findings

- Italy was the most represented birth country in the dataset.
- Height and weight showed virtually no correlation with goals scored (r ≈ 0.06 for both).
- Average matches played peaks in the 21–30 age range.
- xG and actual goals were closely aligned for the highest-xG player (M. Dugandžić: xG 23.23 vs 22 actual goals).

- Goalkeeper-specific metrics (Save rate, xG against, Prevented goals, Exits per 90) were structurally separated from outfield stats, making GK the most cleanly classifiable position category.
- Forwards and midfielders share overlapping statistics (Key passes per 90, xA, Progressive runs), which created classification difficulty in the models.

3 Machine Learning

3.1 Models Applied

The notebook implemented a full range of machine learning models across regression and classification tasks. All models were applied to the same dataset using sklearn pipelines with StandardScaler and OneHotEncoder preprocessing. The train/test split used was 80/20 (or 70/30 in some cells) with random_state=42. The models and their targets were:

Model	Target	Accuracy	Notes
Linear Regression	xA per 90	R^2: 0.89 (pipeline)	Ridge + TruncatedSVD (500 components)
Linear Regression (simple)	xA per 90	R^2: 0.097	Age + Accurate passes % only
Logistic Regression	xA per 90 > 0.5	99.88%	Heavily imbalanced classes — high acc. misleading
Decision Tree	xA per 90 > 0.5	99.85%	Same imbalance issue; macro avg F1 = 0.50
Gradient Boosting	xA per 90 > 0.5	99.88%	Same imbalance; precision on minority class = 0
Random Forest	xA per 90 > 0.5	99.88%	Best ensemble result; same class imbalance noted
XGBoost	Market value	MSE / R^2	Used on market value prediction task
KMeans Clustering	Unsupervised	N/A	Player segmentation by statistics
SVM (SVC)	xA per 90 > 0.5	Tested	Applied as part of model comparison

Note on the accuracy figures: the 99.88% accuracy values for Logistic Regression, Decision Tree, Gradient Boosting, and Random Forest all applied to a binary classification of xA per 90 > 0.5. Because only 4 out of ~16,500 players exceeded this threshold, predicting 'No' for everyone yields 99.88% accuracy. The macro-average F1 score for these models was 0.50, confirming the high accuracy is an artifact of class imbalance, not genuine predictive power on the minority class. This is documented by the UndefinedMetricWarning printed in the notebook output for several models.

3.2 Model Results & Evaluation

3.2.1 Regression — Ridge + TruncatedSVD Pipeline

The best regression result in the notebook was achieved with a pipeline combining:

- SimpleImputer (mean for numerical, most_frequent for categorical)
- StandardScaler on numerical columns
- OneHotEncoder on categorical columns
- TruncatedSVD with 500 components for dimensionality reduction
- Ridge Regression as the final estimator

Result on predicting xA per 90: MSE = 0.00056, R^2 = 0.889. This was the strongest regression result in the notebook.

3.2.2 Regression — Linear Regression (Simple)

A simpler Linear Regression model using only Age and Accurate passes % as features to predict xA per 90 produced: MSE = 0.0053, R^2 = 0.097. The low R^2 reflects that two features alone are not sufficient to explain variation in xA per 90.

3.2.3 Classification Evaluation

Classification models were evaluated using:

- accuracy_score — overall accuracy on the test set
- confusion_matrix — to show true/false positives and negatives per class
- classification_report — precision, recall, F1-score, support, macro average, weighted average

The KNN classifier was also implemented and used as part of the evaluation process. All classification reports and confusion matrices were printed directly in the notebook output cells.

3.2.4 Gradient Descent Implementation

A manual gradient descent implementation was included in the notebook for the market value prediction task. The notebook:

- Loaded the dataset and selected 10 features: Age, Matches played, Minutes played, Goals, Assists, Key passes per 90, Long passes per 90, Dribbles per 90, Shot assists per 90, Interceptions per 90.
- Normalised both X and Y to [0, 1] range using min-max scaling.
- Added a bias column (column of 1s) to X using np.insert.
- Implemented Init_param, cost_fun, and grad_descent_step functions manually with NumPy.
- Ran 1000 iterations with learning rate $\alpha = 0.01$.
- Plotted the cost over iterations, showing convergence.
- Compared manual gradient descent results against sklearn's LinearRegression, which returned coefficients and intercept consistent with the manual implementation.

3.3 MySQL Database Integration

The notebook includes a full MySQL database management script using mysql.connector. The implementation provides a menu-driven command-line interface with the following operations:

- Option 1 — Create a new database: prompts for a name and executes createDB(DbName).
- Option 2 — Check if a database exists: uses checkDB(DbName, cursor) to verify existence.
- Option 3 — List all databases: executes SHOW DATABASES and prints results.
- Option 4 — Fetch all data from a table: connects to database 'ashash', queries table 'ash', and prints all rows.
- Option 0 — Exit: breaks the loop and closes the connection.

The script uses a try/except/finally structure throughout, with mysql.connector.Error handling to catch and report database errors without crashing the program. The connection is explicitly closed in the finally block. Helper functions defined include createDB, checkDB, and getAll.

Note: the database and table names in the implementation are hardcoded as 'ashash' and 'ash' for the data retrieval option. The script does not include REST API endpoints — data access is provided through the direct MySQL connection only.

3.4 Streamlit Visualisation

Two Streamlit applications were implemented in the notebook:

- Simple Calculator app: a Streamlit UI with two number inputs and four operation buttons (Add, Subtract, Multiply, Divide), displaying results with st.success(). A division-by-zero guard was included.
- Team Market Value Analyser: a file uploader widget that accepts a CSV, reads it into a DataFrame, and calls a plot_top_teams_market_values function. This function identifies the top 3 teams by average market value and renders a Seaborn bar plot via Matplotlib. The app was designed to allow NAC Breda analysts to upload updated player data and immediately visualise team-level market value comparisons without writing code.

The Streamlit apps were run from within the notebook environment, producing a warning that they need to be launched with streamlit run for full browser functionality.

3.5 Mathematics for Machine Learning

3.5.1 Linear Algebra — Matrix Properties

The notebook includes a NumPy implementation verifying 13 matrix algebra properties, including commutativity, associativity, distributivity, transpose rules, and inverse rules (e.g. $(AB)^{-1} = B^{-1}A^{-1}$, $(A^T)^{-1} = (A^{-1})^T$). All 13 properties were verified using np.array_equal with randomly generated test matrices.

3.5.2 Normal Equations

The notebook solved two linear systems using np.linalg.solve and derived the least squares estimate $\theta = (X^TX)^{-1}X^Ty$ for a 10-sample regression problem with known data ($y = 2x + 3 +$ noise). The fitted parameters were $\theta_0 \approx 3.068$ and $\theta_1 \approx 1.985$, close to the true values of 3 and 2 respectively.

3.5.3 Symbolic Mathematics with SymPy

The DataLab symbolic mathematics tasks were completed in the notebook using SymPy:

- Task 1: Defined symbolic expressions ex1 = $2x^2 - xy + 3$ and ex2 = $(x \cdot ex1 + 2x + y)/(x^2 + y)$, evaluated ex2 at x=-2, y=1 (result: -5.8), and solved $2x + 5 = 11$ (solution: $x = 3$).
- Solved a 2-variable system: $2x + y = 5$ and $x - 2y = 15$ (solution: $x = 5$, $y = -5$).
- Calculated lim(sin(x)/x) as $x \to 0$ (result: 1).
- Differentiated $f(x) = x^3 + 3x^2 + \sin(x)$ (result: $3x^2 + 6x + \cos(x)$).
- Computed indefinite integral of $x \cdot \sin(x)$ (result: $-x \cdot \cos(x) + \sin(x)$).
- Computed Taylor series expansion of e^x around x=0 up to 4th order.

3.5.4 Calculus — Khan Academy

The Differential Calculus course on Khan Academy was completed and a certificate of completion was submitted. The certificate PDF is available in the GitHub repository at:

https://github.com/BredaUniversityADSAI/2023-24b-fai1-adsai-AshrafNahlouse234669/blob/main/Deliverables%20week%207/CalMacLea_234669.pdf

4 Ethical Considerations

4.1 Three Elements Vital for Ethical Organisational Capacity at NAC

1. Data Privacy: Player data — particularly biometric details, health records, and personal contract information — is sensitive. NAC needs robust access controls, clear data retention policies, and active privacy awareness among staff.
2. Data Quality: Decisions based on incorrect or biased data can harm players' careers. NAC should have processes in place to validate incoming data, flag anomalies, and avoid acting on metrics that are incomplete or unreliable for certain player demographics.
3. Fairness: Algorithmic tools must not systematically disadvantage players based on nationality, physical profile, or other protected characteristics. Fairness requires ongoing auditing of model outputs.

4.2 Responsible Parties within NAC

- Data Privacy: IT and security teams, with oversight from senior management to embed privacy culture across the organisation.
- Data Quality: Data analysts and data scientists, with management accountability for ensuring processes meet ethical standards.
- Fairness: Data scientists designing and evaluating the models, alongside the sporting director who ultimately acts on the outputs.

4.3 How NAC Currently Considers These Elements

Based on available research, NAC Breda does not appear to have disclosed a formal AI ethics policy. However, as a Dutch professional football club, NAC is subject to GDPR and operates within a regulatory environment that mandates baseline data protection. Their published sponsorship relationships and public communications show no evidence of discriminatory practices. The club's use of performance dashboards appears to follow standard industry practice, though without publicly documented auditing procedures.

4.4 Ethical Framework Applied

4.4.1 GDPR Considerations

- Lawful basis for processing: Player statistics gathered for scouting purposes must have a clear legal basis — typically legitimate interest or contractual necessity.
- Data minimisation: Only collecting the statistics actually needed for position prediction, rather than building comprehensive profiles beyond the stated purpose.
- Right of access and correction: Players whose data is processed should be able to access and contest it, particularly where career decisions may follow.
- Data security: Appropriate technical measures (encryption, access logging, role-based permissions) should be in place for any system storing player data.

The model built in this project uses statistical data that can be anonymised rather than directly identifying records, which reduces the GDPR risk profile compared to systems processing personal health or biometric data.

4.4.2 Ethical Guidelines for Statistical Practice

- Integrity: Models that performed worse were not excluded from reporting. The class imbalance issue with the 99.88% accuracy results is explicitly acknowledged rather than presented as a genuine achievement.
- Transparency: Methodology, including preprocessing decisions and model selection rationale, is documented in the notebook with comments at each stage.
- Competence: Models were evaluated using multiple metrics — not just accuracy, but confusion matrices, classification reports, precision, recall, and F1-scores.
- Responsibility: Limitations of the models — including the misleading high accuracy on imbalanced targets — are identified rather than glossed over.

4.5 Ethical Problems Identified within NAC

- Resource proportionality: Comprehensive data security infrastructure carries a real cost. For a club the size of NAC, allocating significant budget to data governance needs to be weighed against core football operations. There is a risk that pursuing best-practice data security could divert resources from training, youth development, or coaching.
- Residual bias: Even with careful data preparation, biases can persist. Players from less-documented leagues may have incomplete statistical records, which could cause the model to underperform for those players and inadvertently disadvantage them in scouting processes.
- Class imbalance risk: The high accuracy results observed in this project's classification models were driven by severe class imbalance, not genuine predictive performance. If decision-makers act on accuracy figures without understanding the underlying distribution, they risk deploying a model that appears to work but fails on the cases that matter.

4.6 Recommendations for NAC

- Balance resources: Implement data governance measures proportionate to the club's size and actual risk exposure.
- Audit for bias: Periodically review model outputs broken down by player nationality, league of origin, and age to check for systematic disparities.
- Train staff: Ensure scouts and coaching staff using model outputs understand what predictions represent and what their limitations are. A model accuracy figure is not a verdict.
- Address class imbalance: For future iterations of the classification models, apply techniques such as SMOTE, class-weighted loss functions, or threshold adjustment to improve minority class recall before deployment.

5 Recommendations

5.1 Looking for Players

The most practical scouting application of this project's models is identifying players who are statistically performing in a profile that does not match their listed position. A player whose statistics align more closely with an attacking role but who is being used as a midfielder may be undervalued — their market value may not yet reflect what they could achieve in the right position.

To find these players:

- Filter the dataset for players aged 17–24 with above-average statistics for their listed position.
- Compare the predicted position from the Random Forest model against their current listed role.
- Where there is a mismatch, cross-reference with platforms like Transfermarkt to assess market value trajectory. Players whose value has dipped recently despite strong stats are worth investigating further.
- Use the Contract Duration Left feature (already calculated in the notebook) to prioritise players whose contracts are expiring — these represent lower acquisition costs.

5.2 Analysing Scouted Players

Once a shortlist is identified, the Ridge regression pipeline ($R^2 = 0.889$ for xA per 90) and the market value modelling can be used to add financial context alongside the positional analysis. For each candidate, the classification model outputs a predicted position category alongside class probabilities — not just 'Midfielder' but the full distribution across all four classes. A player with a high forward probability despite being listed as a midfielder is a stronger candidate than one where probabilities are evenly spread.

5.3 Considerations for Player Acquisition

When the model's predicted position and the player's actual statistical output diverge, that gap is where the potential value lies. Clubs that have deployed a player in a sub-optimal role are often willing to sell at below-peak valuations, particularly if recent performances have been inconsistent. Acquiring the player and repositioning them — guided by the model's prediction — carries a reasonable evidence base.

That said, model predictions should always be one input among several. Physical and medical assessment, contract situation, and cultural fit within the squad all matter. The model narrows the search space and flags candidates worth a closer look; it does not replace the judgement calls that are ultimately required.

6 Referencing

6.1 Use of Generative AI

Generative AI (ChatGPT, OpenAI) was used at three points during this project:

- Report title and subtitle: After the report content was complete, a prompt was used to generate a creative title. Prompt: 'Give me a title of a report written about a predictive machine learning model on football data' (24-01-2024).
- Text expansion: Source paragraphs written by the author were expanded for detail using the prompt 'Please broaden the following text:' (22-01-2024).
- Professional tone: Drafted sections were refined with the prompt 'Make this text a little bit more professional:' (22-01-2024).

In all cases, the original ideas, analysis, and conclusions were authored independently. AI was used only for language and presentation, not for generating findings or analytical content.

6.2 Webpages

- Transfermarkt. (n.d.). Football transfers, rumours, market values and news. https://www.transfermarkt.com/

6.3 Scholarly Sources

- Velasquez, M. G. (2017). Business Ethics: Concepts and Cases. Pearson.
- Twin, A. (2023, March 17). Business Ethics: Definition, Principles, Why They're Important. Investopedia. https://www.investopedia.com/terms/b/business-ethics.asp
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